Fundamentals of Machine learning

What is machine Learning?

- Machine Learning is the science (and art) of programming computers so they can learn from data.

- Machine Learning is a field of study that gives computers the ability to learn without being explicitly programmed.

- Machine Learning is the use and development of computer systems that can learn and adapt without following explicit instructions, by using algorithms and statistical models to analyse and draw inferences from patterns in data.

Definitions of Machine Learning

- ► Machine learning (ML) is a type of artificial intelligence (AI) that allows software applications to become more accurate at predicting outcomes without being explicitly programmed to do so. Machine learning algorithms use historical data as input to predict new output values.

- ► Machine Learning is a branch of Artificial Intelligence that allows machines to learn and improve from experience automatically. It is defined as the field of study that gives computers the capability to learn without being explicitly programmed. It is quite different than traditional programming.

Machine learning types

Machine learning are classified into the fallowing types

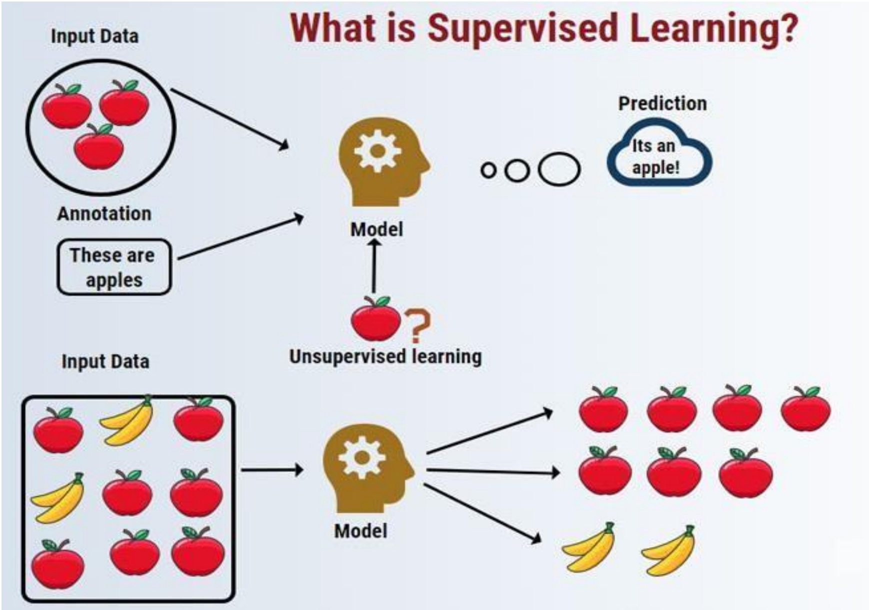
1. **Supervised Learning**
2. **Unsupervised Learning**
3. **Semi- supervised Learning**
4. **Reinforcement Learning**

Supervised Learning:

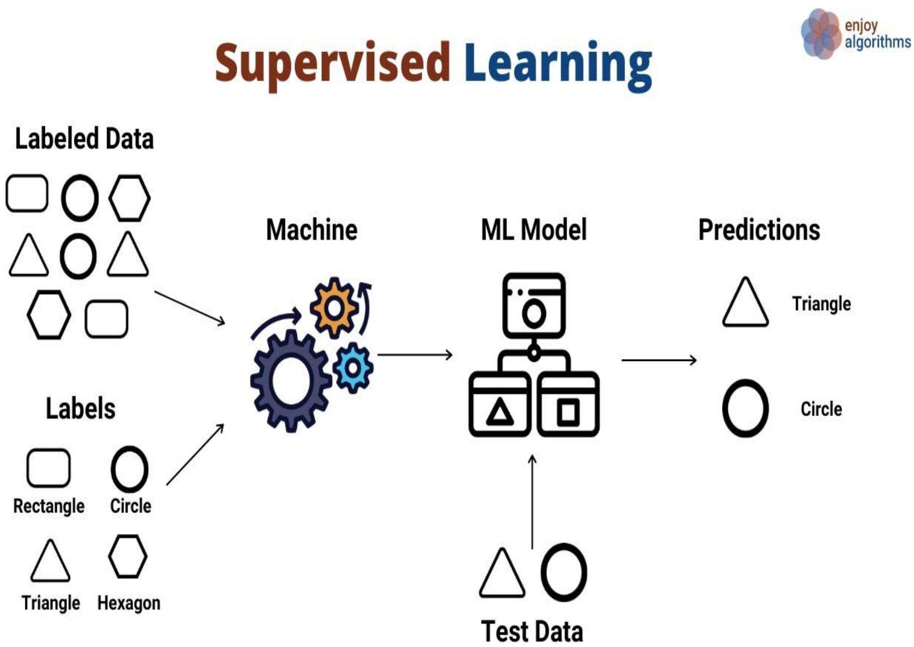
1. Supervised Learning is a type of machine learning algorithms which learns from the labelled dataset. This machine learning method that needs supervision similar to the student-teacher relationship.
2. In supervised learning, the training data you feed to the algorithm includes the desired solutions, called labels.
3. In supervised Learning, a machine is trained with well-labeled data, which means some data is

already tagged with correct outputs.

4. So, whenever new data is introduced into the system, supervised learning algorithms analyze this sample data and predict correct outputs with the help of that labeled data.



From the above figure we can notice that machine is trained with well labelled dataset. Which includes the features o the apple. hence when an apple is given as an input the model is able to predict it as an apple.



It is classified into two different categories of algorithms. These are as follows:

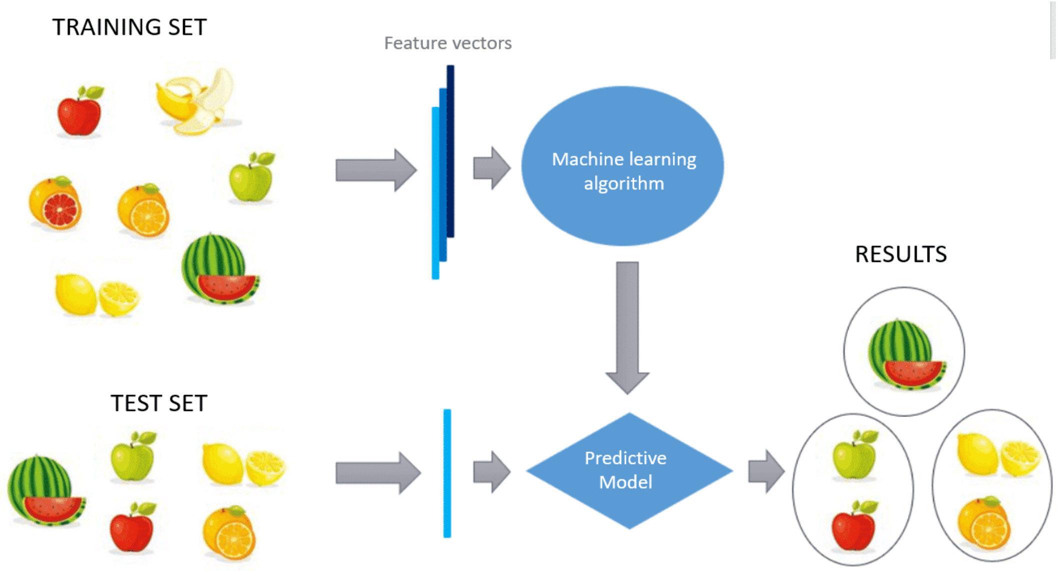
- Classification: It deals when output is in the form of a category such as Yellow, blue, right, or wrong, etc.
- Regression: It deals when output variables are real values like age, height, etc.

This technology allows us to collect or produce data output from experience. It works the same way as humans learn using some labeled data points of the training set. It helps in optimizing the performance of models using experience and solving various complex computation problems.

Unsupervised Learning:

1. Unlike supervised learning, unsupervised Learning does not require classified or well-labeled data to train a machine.

2. It aims to make groups of unsorted information based on some patterns and differences even without any labelled training data. In unsupervised Learning, no supervision is provided, so no sample data is given to the machines.
3. Hence, machines are restricted to finding hidden structures in unlabeled data by their own.

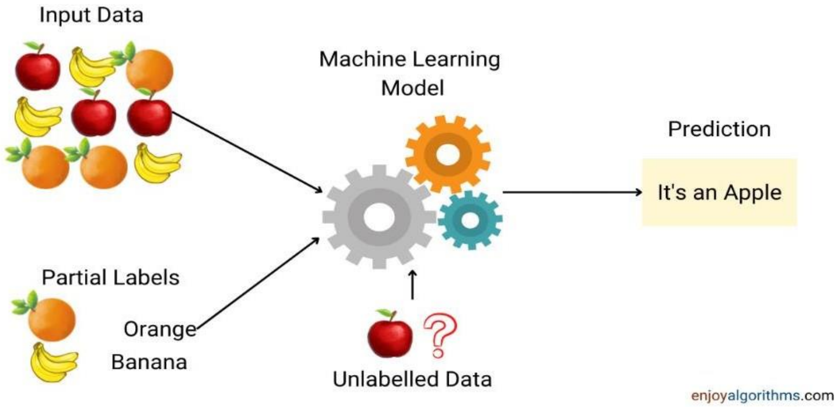


It is classified into two different categories of algorithms. These are as follows:

- Clustering: It deals when there is a requirement of inherent grouping in training data, e.g., grouping students by their area of interest.
- Association: It deals with the rules that help to identify a large portion of data, such as students who are interested in ML and interested in AI.

Semi- supervised Learning:

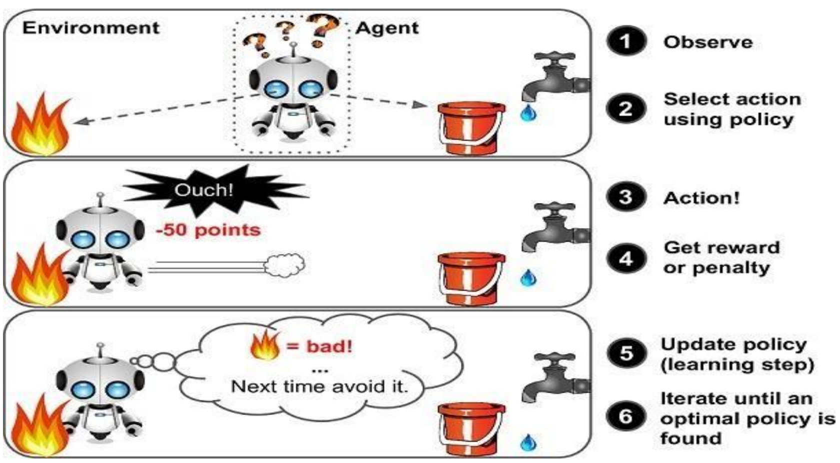
1. Semi supervised learning algorithms can deal with partially labeled training data, usually a lot of unlabeled data and a little bit of labeled data. This is called semi supervised learning
2. Some photo-hosting services, such as Google Photos, are good examples of this. Once you upload all your family photos to the service, it automatically recognizes that the same person A shows up in photos 1, 5, and 11, while another person B shows up in photos 2, 5, and 7.
3. This is the unsupervised part of the algorithm (clustering). Now all the system needs are for you to tell it who these people are. Just one label per person, 4 and it can name everyone in every photo, which is useful for searching photos



Reinforcement Learning

1. Reinforcement Learning is a very different type of learning. The learning system, is called an agent in this context, can observe the environment, select, and perform actions, and get rewards in return (or penalties in the form of negative rewards).

2. It must then learn by itself what is the best strategy, called a policy, to get the most reward over time. A policy defines what action the agent should choose when it is in a given situation.

3. 

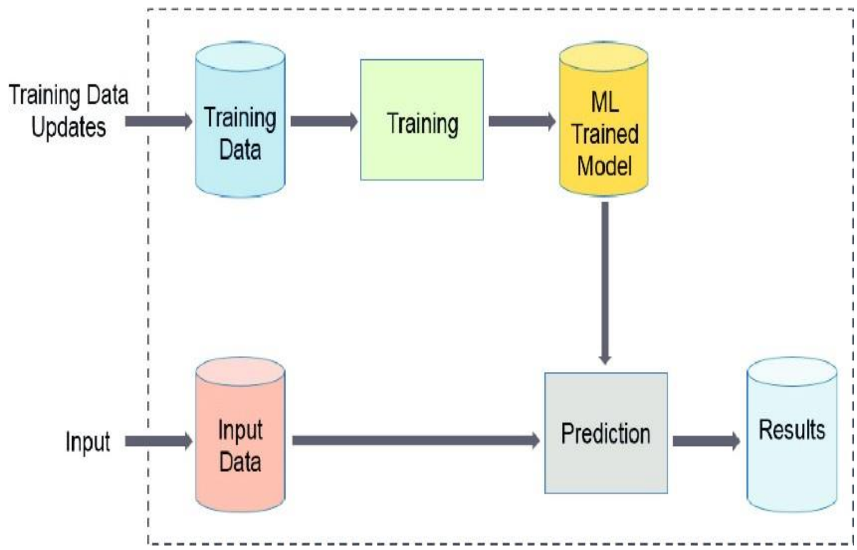
Differentiate between supervised machine learning and Unsupervised machine learning

Supervised Learning	Unsupervised Learning
1. Supervised learning algorithms are trained using labelled data.	1. Unsupervised learning algorithms are trained using unlabelled data.
2. Supervised learning model takes direct feedback to check if it is predicting correct output or not	2. Unsupervised learning model does not take any feedback.
3. Supervised learning model predicts the output.	3. Unsupervised learning model finds the hidden patterns in data.
4. In supervised learning, input data is provided to the model along with the output.	4. In unsupervised learning, only input data is provided to the model.
5. The goal of supervised learning is to train the model so that it can predict the output when it is given new data.	5. The goal of unsupervised learning is to find the hidden patterns and useful insights from the unknown dataset.
6. Supervised learning needs supervision to train the model.	6. Unsupervised learning does not need any supervision to train the model.
7. Supervised learning can be categorized in Classification and Regression problems.	7. Unsupervised Learning can be classified in Clustering and Associations problems.
8. Supervised learning can be used for those cases where we know the input as well as corresponding outputs.	8. Unsupervised learning can be used for those cases where we have only input data and no corresponding output data.
9. It includes various algorithms such as Linear Regression, Logistic Regression, Support Vector Machine, Multi-class Classification, Decision tree, etc.	9. It includes various algorithms such as Clustering, KNN, and Apriori algorithm.
10. Supervised learning model produces an accurate result.	10. Unsupervised learning model may give less accurate result as compared to supervised learning

Machine learning workflow

Machine learning workflows define which phases are implemented during a machine learning project. The typical phases include

1. data collection,
2. data pre-processing,
3. building datasets,
4. model training and
5. refinement, evaluation,
6. and deployment to production



Machine Learning Workflow

Machine learning applications

1. Image Recognition:

It is used to identify objects, persons, places, digital images, etc. The popular use case of image recognition and face detection is, **Automatic friend tagging.**

2. Speech Recognition

Speech recognition is a process of converting voice instructions into text, and it is also known as "**Speech to text**", or "**Computer speech recognition**." At present, machine learning algorithms are widely used by various applications of speech recognition. **Google assistant**, **Siri**, **Cortana**, and **Alexa** are using speech recognition technology to follow the voice instructions.

3. Traffic prediction:

If we want to visit a new place, we take help of Google Maps, which shows us the correct path with the shortest route and predicts the traffic conditions. It predicts the traffic conditions such as whether traffic is cleared, slow-moving, or heavily congested with the help of two ways:

- **Real Time location** of the vehicle form Google Map app and sensors
- **Average time has taken** on past days at the same time.

4. Product recommendations:

Machine learning is widely used by various e-commerce and entertainment companies such as **Amazon**, **Netflix**, etc., for product recommendation to the user. Whenever we search for some product on Amazon, then we started getting an advertisement for the same product while internet surfing on the same browser and this is because of machine learning.

5. Self-driving cars:

One of the most exciting applications of machine learning is self-driving cars. Machine learning plays a significant role in self-driving cars. Tesla, the most popular car manufacturing company is working on self-driving car. It is using unsupervised learning method to train the car models to detect people and objects while driving.

6. Email Spam and Malware Filtering:

Whenever we receive a new email, it is filtered automatically as important, normal, and spam. We always receive an important mail in our inbox with the important symbol and spam emails in our spam box, and the technology behind this is Machine learning. Below are some spam filters used by Gmail:

- Content Filter
- Header filter
- General blacklists filter
- Rules-based filters
- Permission filters

8. Online Fraud Detection:

Machine learning is making our online transaction safe and secure by detecting fraud transaction. Whenever we perform some online transaction, there may be various ways that a fraudulent transaction can take place such as **fake accounts**, **fake ids**, and **steal money** in the middle of a transaction. So, to detect this, **Feed Forward Neural network** helps us by checking whether it is a genuine transaction or a fraud transaction.

10. Medical Diagnosis:

In medical science, machine learning is used for diseases diagnoses. With this, medical technology is growing very fast and able to build 3D models that can predict the exact position of lesions in the brain. It helps in finding brain tumors and other brain-related diseases easily.

Challenges in ML

1. ***Data Collection***: Data plays a key role in any use case. 60% of the work of a data scientist lies in collecting the data. Once the data is collected, structure the data and store it in the database.
2. ***Less Amount of Training Data :*** Once the data is collected it has to be validate if the quantity is sufficient for the use case
3. ***Non-representative Training Data :*** The training data should be representative of the new cases to generalize well i.e., the data we use for training should cover all the cases that occurred and that is going to occur. By using a non-representative training set, the trained model is not likely to make accurate predictions.
4. ***Poor Quality of Data:*** Noisy data, incomplete data, inaccurate data, and unclean data lead to less accuracy in classification and low-quality results. Hence, data quality can also be consideredas a major common problem while processing machine learning algorithms.
5. ***Irrelevant/Unwanted Features*** If the training data contains a large number of irrelevant features and enough relevant features, the machine learning system will not give the results as expected
6. ***Overfitting the Training Data*** Whenever a machine learning model is trained with a huge amount of data, it starts capturing noise and inaccurate data into the training data set. It negatively affects the performance of the model.
7. ***Underfitting the Training data*** Underfitting is just the opposite of overfitting. Whenever a machine learning model is trained with fewer amounts of data, and as a result, it provides incomplete and inaccurate data and destroys the accuracy of the machine learning model.

Exams answer

1. Inadequate Training Data and Poor quality of data

The major issue that comes while using machine learning algorithms is the lack of quality as well as quantity of data. Although data plays a vital role in the processing of machine learning algorithms, many data scientists claim that inadequate data, inaccurate data, noisy data, and unclean data are extremely exhausting the machine learning algorithms. This leads to less accuracy in classification and low-quality results. Hence, data quality can also be considered as a major common problem while processing machine learning algorithms.

2. *Non-representative training data*

To make sure our training model is generalized well or not, we have to ensure that sample training data must be representative of new cases that we need to generalize. The training data must cover all cases that are already occurred as well as occurring.

3. *Overfitting*

Overfitting is one of the most common issues faced by Machine Learning engineers and data scientists. Whenever a machine learning model is trained with a huge amount of data, it starts capturing noise and inaccurate data into the training data set. It negatively affects the performance of the model. The main reason behind overfitting is using non-linear methods used in machine learning algorithms as they build non-realistic data models.

4. *Underfitting:*

Underfitting is just the opposite of overfitting. Whenever a machine learning model is trained with fewer amounts of data, and as a result, it provides incomplete and inaccurate data and destroys the accuracy of the machine learning model.

5. *Monitoring and maintenance*

as we know that generalized output data is mandatory for any machine learning model; hence, regular monitoring and maintenance become compulsory for the same. Different results for different actions require data change; hence editing of codes as well as resources for monitoring them also become necessary.

6. *Lack of skilled resources*

Although Machine Learning and Artificial Intelligence are continuously growing in the market, still these industries are fresher in comparison to others. The absence of skilled resources in the form of manpower is also an issue. Hence, we need manpower having in-

Deployment: Deploying ML models in production environments can be difficult, as it requires expertise in both ML and software engineering.

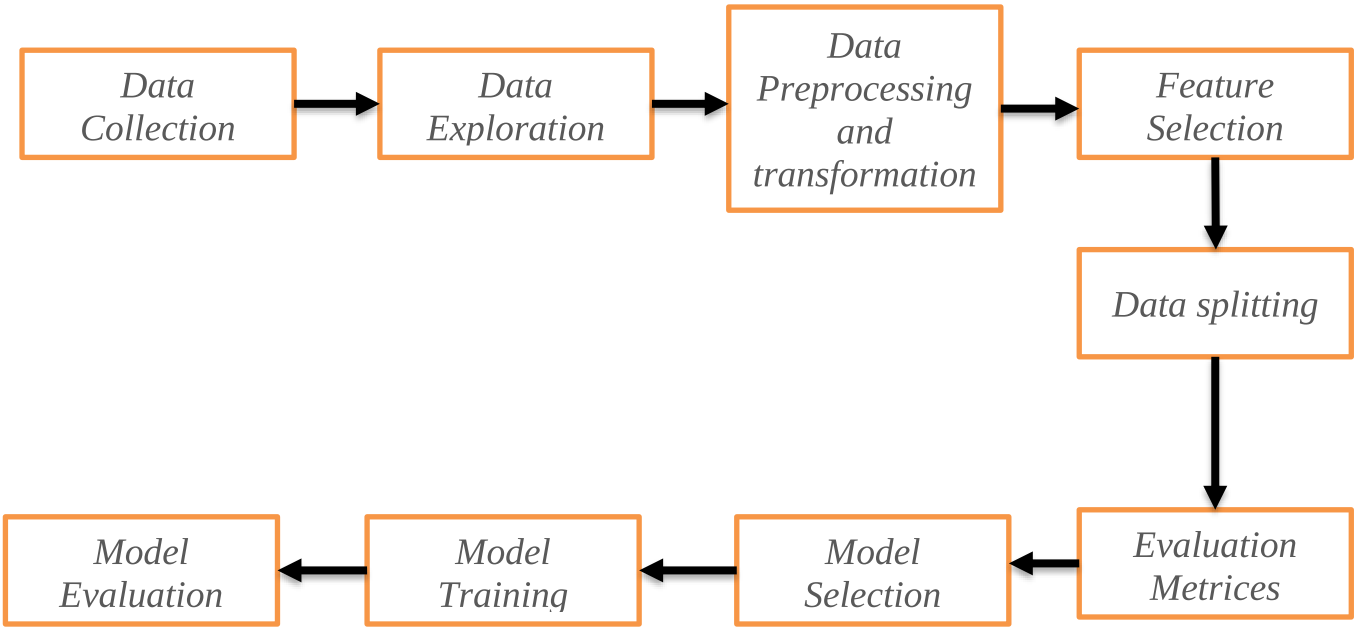
7. *Model Selection:*

There are many different ML algorithms to choose from, and selecting the right one for a particular problem is challenging.

8. *Feature Engineering:*

Extracting useful features from raw data is a critical step in ML, but it can be difficult to identify the most relevant features.

Building a model



The various Steps involved in building machine Learning model

1. **Data Exploration:** The first step is to explore the data and understand the characteristics of the dataset. This includes understanding the number of observations and variables, the data types of each variable, and the distribution of the data. This can be done by using summary statistics and visualizations such as histograms, box plots, and scatter plots.

2. **Data Cleaning:** The next step is to clean the data. This includes handling missing or corrupted data, removing outliers, and addressing any other data quality issues. This step is important because dirty data can lead to inaccurate or unreliable models.

3. **Data Transformation**: After cleaning the data, it may be necessary to transform the data to make it suitable for the machine learning model. This can include normalizing the data, scaling the data, or creating new variables.

4. **Feature Selection:** Once the data is cleaned and transformed, it is important to select the relevant features that will be used to train the model. This step can be done by using techniques such as correlation analysis, principal component analysis, or mutual information.

5. **Data Splitting**: The next step is to split the data into training, validation, and test sets. The training set is used to train the model, the validation set is used to tune the model's parameters, and the test set is used to evaluate the model's performance.

6. **Feature Engineering:** This step is to create new features that will be useful in the model. This can include creating interaction terms, polynomial terms, or binning variables.

7. **Evaluation Metric:** Selecting the right evaluation metric will help to evaluate the model's performance. Common evaluation metrics include accuracy, precision, recall, F1 score, and area under the ROC curve.

8. **Model Selection:** After the data is prepared, the next step is to select the appropriate machine learning model. This can be done by comparing the performance of different models using the evaluation metric.

9. **Model Training:** Once the model is selected, it is trained using the training dataset.

10. **Model Evaluation:** Finally, the model's performance is evaluated using the test dataset and the evaluation metric selected.

What is Data Science?

- Data science is the study of data to extract meaningful insights for business. It is a multidisciplinary approach that combines principles and practices.

- Data science is an interdisciplinary field that uses scientific methods, processes, algorithms, and systems to extract knowledge and insights from noisy, structured, and unstructured data, and apply knowledge from data across a broad range of application domains

How Data Science works?

Data science uses techniques such as machine learning and artificial intelligence to extract meaningful information and to predict future patterns and behaviors. Advances in technology, the internet, social media, and the use of technology have all increased access to big data.

What do data scientist do?

- “More generally, a data s

- cientist is someone who knows how to extract meaning from and interpret data, which requires both tools and methods from statistics and machine learning, as well as being human.
- She spends a lot of time in the process of collecting, cleaning, and munging data, because data is never clean.
- This process requires persistence, statistics, and software engineering skills—skills that are also necessary for understanding biases in the data, and for debugging logging output from code.

Data Science uses

Healthcare Sector

- Data science helps in the interpretation of medical images such as X Rays, MRIs, CT scans, etc. If the medical history of a patient is known, data science can predict future medical health.
- Diagnosis of various diseases like cancer, schizophrenia, Alzheimer's, etc can be predicted with help of pattern matching and spectrum analysis. It can even provide a better understanding of genetic tissues and the reaction to specific drugs or diseases.

Tourism Industry

- Data Science can give enhanced experience in tourism as well. It can predict flight delays in advance and send messages to the passengers beforehand. It can track any deals or offers on the dynamic pricing of hotels and flights and timely notify the customers about those deals

Finance and Insurance Sector

- Data Science algorithms can help prevent fraud in the finance sector. It can check the whole financial history and present situation of a person to predict if he would be able to pay the debt thus minimizing the risk of any financial losses or debts.

Important questions?

1.what is machine learning? what are the types of ML? list the challenges of ML?

2. list the applications of ML?

3.what are the steps involved in building machine learning model?

4.what is data science? what are uses and applications of data science ?

Introduction to Cloud Computing

Essentials of Cloud Computing

What is Cloud

A cloud refers to a distinct IT environment that is designed for the purpose of remotely provisioning scalable and measured IT resources.

The symbol used to denote the boundary of a cloud environment is as shown in fig. 1.

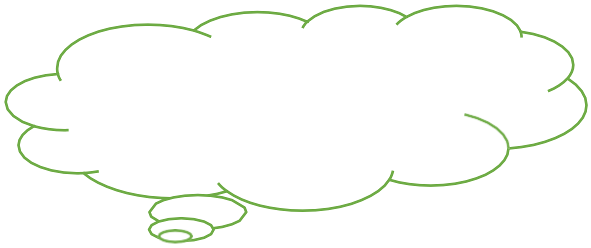


Figure 1. The symbol used to denote the boundary of a cloud environment.

Challenges with corporate data center environment

1.Capital Expenditure

In traditional data center, applications are hosted on on-premise data center. Huge capital investment is required to build the data center. Funds used by a company to acquire, upgrade, and maintain physical assets such as property, plants, buildings, technology, or equipment is high.

2.Time to setup the infrastructure

The time taken to setup the infrastructure on premise is high.

3.Maintenance overhead

The enterprise needs to maintain the infrastructure to provide high performance to the clients which consumes much cost.

4.Resource under/over utilization

Sometimes Resources are not fully utilized and sometimes the resources are over utilized in traditional on-premise data center.

5.Reduced ROI

Maintenance overhead, Resource over/under utilization reduces Return on investment.

6.Security Risks:

On-prem servers are physical assets, and face the same dangers the rest of the building might be exposed to, including fires, floods, or break-ins.

Benefits of Cloud Computing

1. ***Pay as you go pricing model***
   On-demand access to pay-as-you-go computing resources on a short-term basis (such as processorsby the hour), and the ability to release these computing resources when they are no longer needed.

2. ***Lower Capital Expenditure***
   Cloud computing requires lower initial capital investment and users pay only for capacity used.

3. ***Secure Storage Management***
   Various Encryption methods are used to provide security to the data stored in cloud.

4. ***Device and Location Independent***
   Cloud resources can be accessed through internet from anywhere anytime.

5. ***Scalability.***
   By providing pools of IT resources, clouds can instantly and dynamically allocate IT resources to cloud consumers, on-demand This empowers cloud consumers to scale their cloud-based IT resources to accommodate processing fluctuations and peaks automatically or manually

6. ***Highly automated***
   The power of automation in cloud is increased to eliminate the human error.

Features of Cloud Computing

The following six specific characteristics are common to the majority of cloud environments:

1. ***On-demand usage***
Resources are provided on demand from cloud.

2. ***Ubiquitous access***
Cloud resources can be accessed by anywhere anytime.

3. ***Multitenancy (and resource pooling)***
It means that the Cloud provider pulled the computing resources to provide services to multiple customers with the help of a multi-tenant model. There are different physical and virtual resources assigned and reassigned which depends on the demand of the customer.The customer generally has no control or information over the location of the provided resources but is able to specify location at a higher level of abstraction

4. ***Elasticity***

Elasticity is the automated ability of a cloud to transparently scale IT resources, as required in responseto runtime conditions or as pre-determined by the cloud consumer or cloud provider

5. ***Measured usage***

The measured usage characteristic represents the ability of a cloud platform to keep track of the usage of its IT resources, primarily by cloud consumers. Based on what is measured, the cloud provider can charge a cloud consumer only for the IT resources actually used.

6. ***Resiliency***

Resiliency can refer to redundant IT resources within the same cloud (but in different physical locations)or across multiple clouds.

Cloud Deployment Models

A cloud deployment model represents a specific type of cloud environment, primarily distinguished byownership, size, and access.

There are four common cloud deployment models:

- ***Public cloud***
- ***Community cloud***
- ***Private cloud***
- ***Hybrid cloud***

Public Cloud

1.Public Cloud The public cloud makes it possible for anybody to access systems and services. The public cloud may be less secure as it is open to everyone. The public cloud is one in which cloud infrastructure services are provided over the internet to the general people or major industry groups.

The infrastructure in this cloud model is owned by the entity that delivers the cloud services, not by the consumer. It is a type of cloud hosting that allows customers and users to easily access systems and services. This form of cloud computing is an excellent example of cloud hosting, in which service providers supply services to a variety of customers. In this arrangement, storage backup and retrieval services are given for free, as a subscription, or on a per-user basis.

Example: Google App Engine etc.

Advantages of Public Cloud Model:

- Minimal Investment: Because it is a pay-per-use service, there is no substantial upfront fee, making it excellent for enterprises that require immediate access to resources.
- No setup cost: The entire infrastructure is fully subsidized by the cloud service providers, thus there is no need to set up any hardware.
- Infrastructure Management is not required: Using the public cloud does not necessitate infrastructure management.
- No maintenance: The maintenance work is done by the service provider (Not users).
- Dynamic Scalability: To fulfil company's needs, on-demand resources are accessible.

Disadvantages of Public Cloud Model:

- Less secure: Public cloud is less secure as resources are public so there is no guarantee of highlevel security.
- Low customization: It is accessed by many public so it can't be customized according to personal requirements

Private Cloud

The private cloud deployment model is the exact opposite of the public cloud deployment model. It's a one-on-one environment for a single user (customer). There is no need to share your hardware with anyone else. The distinction between private and public clouds is in how you handle all of the hardware. It is also called the "internal cloud" & it refers to the ability to access systems and services within a given border or organization. The cloud platform is implemented in a cloud-based secure environment that is protected by powerful firewalls and under the supervision of an organization's IT department. The private cloud gives greater flexibility of control over cloud resources.

Advantages of Private Cloud Model:

- ***Better Control:*** You are the sole owner of the property. You gain complete command over service integration, IT operations, policies, and user behaviour.
- ***Data Security and Privacy:*** It's suitable for storing corporate information to which only authorized staff have access. By segmenting resources within the same infrastructure, improved access and security can be achieved.
- ***Supports Legacy Systems:*** This approach is designed to work with legacy systems that are unable to access the public cloud.
- ***Customization:*** Unlike a public cloud deployment, a private cloud allows a company to tailor its solution to meet its specific needs.

Disadvantages of Private Cloud Model:

- ***Less scalable:*** Private clouds are scaled within a certain range as there is a smaller number of clients.
- ***Costly:*** Private clouds are costlier as they provide personalized facilities.

Hybrid Cloud

By bridging the public and private worlds with a layer of proprietary software, hybrid cloud computing gives the best of both worlds. With a hybrid solution, you may host the app in a safe environment while taking advantage of the public cloud's cost savings. Organizations can move data and applications between different clouds using a combination of two or more cloud deployment methods, depending on their needs.

Advantages of Hybrid Cloud Model:

- ***Flexibility and control:*** Businesses with more flexibility can design personalized solutions that meet their particular needs.
- ***Cost:*** Because public clouds provide scalability, you'll only be responsible for paying for the extra capacity if you require it.
- ***Security:*** Because data is properly separated, the chances of data theft by attackers are considerably reduced.

Disadvantages of Hybrid Cloud Model:

- ***Difficult to manage:*** Hybrid clouds are difficult to manage as it is a combination of both public and private cloud. So, it is complex.
- ***Slow data transmission:*** Data transmission in the hybrid cloud takes place through the public cloud so latency occurs.

Cloud Service Models

A cloud service model represents a specific, pre-packaged combination of IT resources offered by a cloudprovider.

Three common cloud service models have become widely established and formalized:

- ***Infrastructure-as-a-Service (IaaS)***
- ***Platform-as-a-Service (PaaS)***
- ***Software-as-a-Service (SaaS)***

Infrastructure as a Service (IaaS)

1. It provides access to virtualized cloud infrastructure which comprises of Compute, Storage, and network resources.

2. The general purpose of an IaaS environment is to provide cloud consumers with a high level of control and responsibility over its configuration and utilization.

3. The IT resources provided by IaaS are generally not pre-configured, placing the administrative responsibility directly upon the cloud consumer.

4. This model is therefore used by cloud consumers that require a high level of control over the cloud- based environment they intend to create.

Example of IaaS services are:

1. Elastic Compute Cloud (EC2) from Amazon Web Services (AWS)
2. Virtual machines from Microsoft Azure
3. Google Compute Engine (GCE) from Google Cloud Platform

Platform as a Service (PaaS)

1. The PaaS delivery model represents a pre-defined “ready-to-use” environment typically comprisedof already deployed and configured IT resources.
2. PaaS cloud computing platform is created for the programmer to develop, test, run, andmanage the applications.

Characteristics of PaaS

There are the following characteristics of PaaS -

- Accessible to various users via the same development application.
- Integrates with web services and databases.
- Builds on virtualization technology, so resources can easily be scaled up or down as perthe organization's need.
- Support multiple languages and frameworks.
- Provides an ability to "**Auto-scale**".

Example of PaaS services are:

1. App Cloud from Salesforce.com
2. Google App Engine(GAE) from Google Cloud Platform
3. OpenShift from Red Hat Inc.
4. Oracle cloud platform from Oracle

Software as a Service (SaaS)

SaaS is also known as "**on-demand software**". It is a software in which the applications are hosted by a cloud service provider. Users can access these applications with the help of internet connection and web browser.

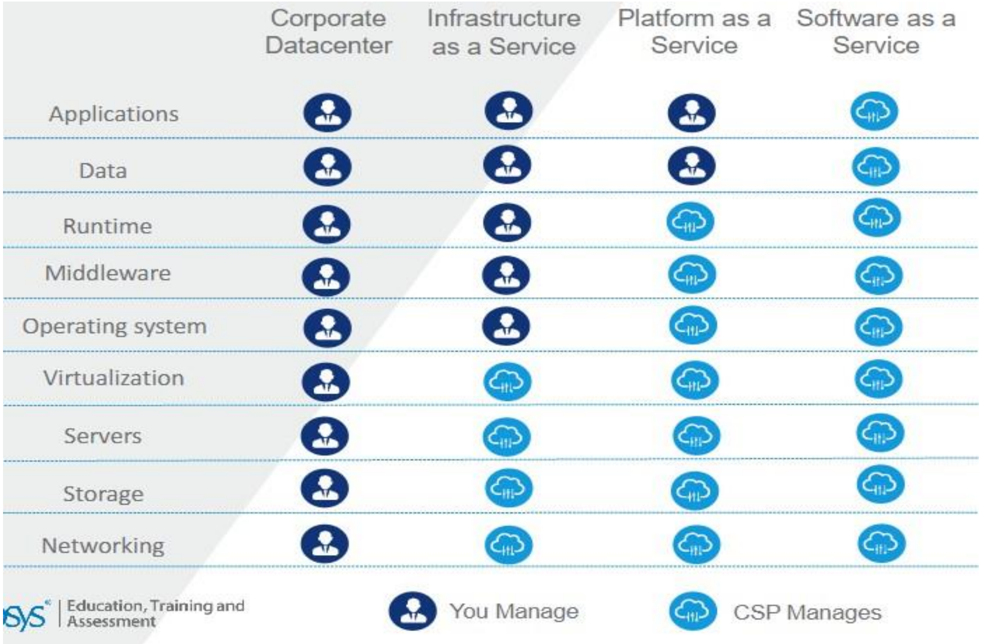
Characteristics of SaaS

There are the following characteristics of SaaS -

- Managed from a central location
- Hosted on a remote server
- Accessible over the internet
- Users are not responsible for hardware and software updates. Updates are applied automatically.
- The services are purchased on the pay-as-per-use basis

Example: BigCommerce, Google Apps, Salesforce, Dropbox, ZenDesk, Cisco WebEx,ZenDesk, Slack, and GoToMeeting.

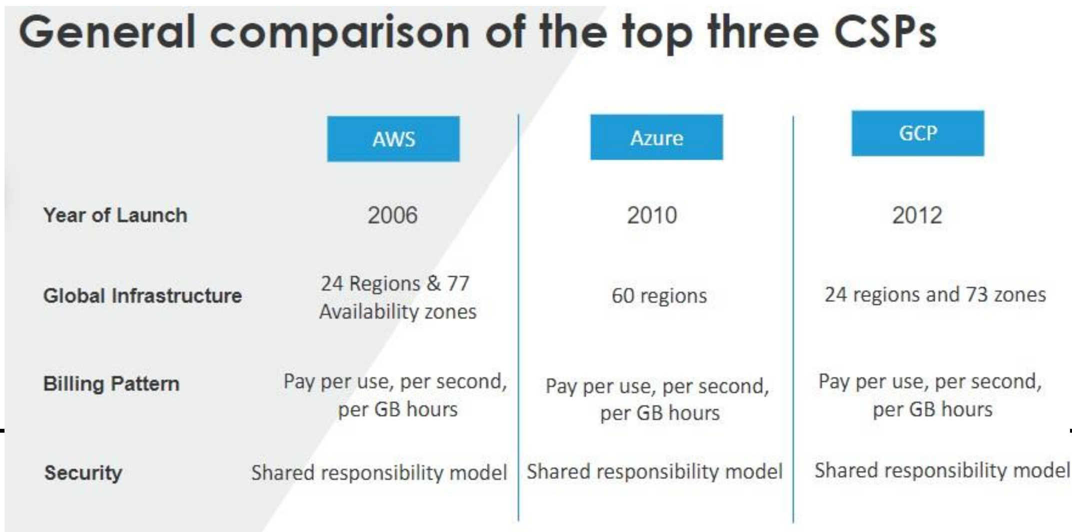
Comparison of cloud service models.



Serverless services

1. It is more suitable for handling asynchronous events.
2. Serverless services will be in active till user invokes it.
3. It allows you to focus on productive business requirements rather than IT infrastructure setup andmanagement.
4. Capacity provisioning and scalability is managed by cloud service provider. Hence it reduces TotalCost of Owner.
5. It allows high availability and fault tolerant to the architecture. It is also known as function as aservice.

Major Cloud service providers


General comparison of the top three CSPs

	AWS	Azure	GCP
Year of Launch	2006	2010	2012
Global Infrastructure	24 Regions & 77 Availability zones	60 regions	24 regions and 73 zones
Billing Pattern	Pay per use, per second, per GB hours	Pay per use, per second, per GB hours	Pay per use, per second, per GB hours
Security	Shared responsibility model	Shared responsibility model	Shared responsibility model

Virtualization

1. Virtualization is the process of converting a physical IT resource into a virtual IT resource.
2. The first step in creating a new virtual server through virtualization software is the allocation of physical IT resources, followed by the installation of an operating system.
3. Virtual servers use their own guest operating systems, which are independent of the operating system in which they were created.
4. Both the guest operating system and the application software running on the virtual server are unaware of the virtualization process, meaning these virtualized IT resources are installed andexecuted as if they were running on a separate physical server.
5. Hypervisor manages resources for Virtual Machines.

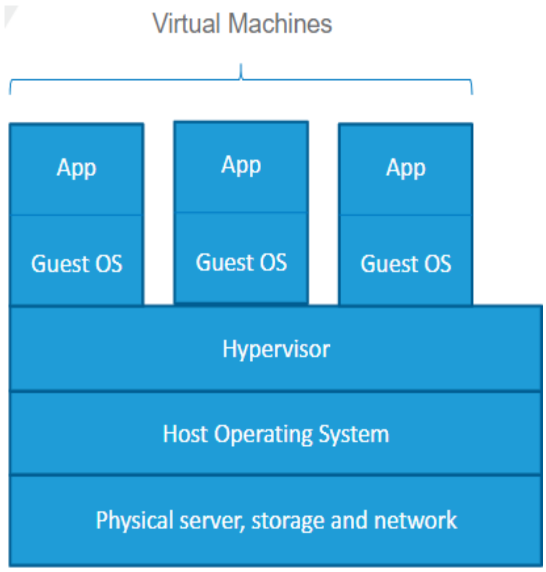
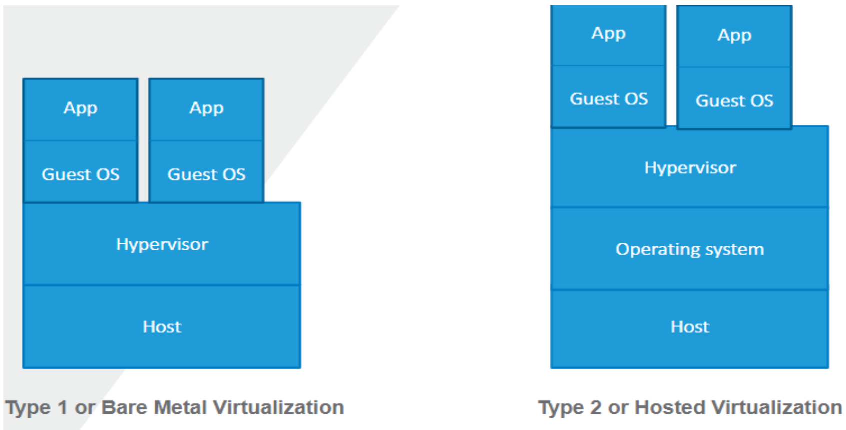


Fig 2: Virtualization

Types of Virtualizations



Benefits of Virtualized Infrastructure

• Cost savings
• Agility
• Reduced downtime

Explore the cloud service providers and services offered by them

1. Amazon Web Services

AWS is a subsidiary of Amazon that provides on-demand cloud computing services to its users.Given below is the list of services offered by AWS:

Storage: - Amazon Simple Storage Service (S3) provides scalable data storage with backup and replication. Compute: - Amazon Elastic Compute Cloud (EC2) provides virtual servers or instances for computing. It isauto-scalable as per the requirement.

Database: - Amazon Relational Database Service provides a fully managed database service that includesOracle, SQL, MySQL, etc

Developer Tool: - AWS CodeCommit provides fully managed private GIT repositories to store code and manage versions. Apart from these services, AWS provides AWS CodePipeline, AWS CodeBuild, AWS CodeDeploy, AWS CLoud9 to support development and deployment.

Machine Learning: - Amazon SageMaker provides services to quickly build, train and deploy models at a big scale. AWS offers ML service for speech recognition, language translation, chatbots, and many other scenarios, with high speed and scalability.

2. Azure

Some of the popular services of Azure cloud provider are:

Azure Active Directory: - Azure Active Directory (AD) is one of the most popular cloud computing servicesfrom Microsoft Azure. Belonging to the Identity section, it is a universal identity platform to ensure the management and security of identities.

Azure CDN: - Its server is designed in a way that it can integrate a lot of storage space, web apps, and Azurecloud services. This is why Azure CDN is used to deliver content securely all across the world.

Azure Data Factory: - Azure Data Factory ingests data from several sources to automate data transmissionand movement. Azure Data Factory creates and monitors workflows.

Azure SQL: - Azure SQL database adds a readily available data storage facility with enhanced performancefor enterprises.

Azure Backup: - Azure Backup allows simple data protection tools from the Azure Web app services, tokeep your data protected from ransomware or loss of any kind.

Containers

1. Containers provide light weight runtime environment for app deployment.
2. It bundles its applications with all the required dependencies, along with its configuration in a singleimage.
3. It uses components of host kernel from the host operating system in order to provide deploymentenvironment to deploy applications.

Benefits of Containers

1. **Rapid Scalability**: As containers boot up time is about fraction of seconds; they provide rapidscalability.
2. **Uses less resources:** Each container does not have a guest operating system. As a result, it consumesvery less resources like storage and memory.
3. **Greater efficiency:** Greater efficiency is achieved as the container requires less resources to spin up.
4. **Increased portability:** Containers are highly portable and can run on any infrastructure.

Containers and Virtual Machines

Containers	Virtual Machines
Boots in milliseconds	Boots in minutes
Requires less resources such as memory, storage etc.,	Needs more resources
Higher level isolation	Lower level isolation
Highly portable	Highly portable
Shares OS libraries and Kernel	Doesn't share OS with the host machine
Doesn't need a hypervisor	Requires a hypervisor

Containerization using Docker

• Open-source Docker Engine.
• Provide both Linux and Windows based containers
• Containerized applications run anywhere consistently
• Docker CLI, API
• Docker Image
• Docker File

Cloud Billing

A Cloud Billing account defines who pays for a given set of Google Cloud resources. To use Google Cloud services, you must have a valid Cloud Billing account, and must link it to your Google Cloud projects. Your project's Google Cloud usage is charged to the linked Cloud Billing account.

You must have a valid Cloud Billing account even if you are in your free trial period or if you only use Google Cloud resources that are covered by the Google Cloud Free Tier.

SLA

A **Service Level Agreement (SLA)** is the bond for performance negotiated between the cloud services provider and the client. Earlier, in cloud computing all Service Level Agreements were negotiated between a client and the service consumer. Nowadays, with the initiation of large utility-like cloud computing providers, most Service Level Agreements are standardized until a client becomes a large consumer of cloud services. Service level agreements are also defined at different levels which are mentioned below:

- Customer-based SLA
- Service-based SLA
- Multilevel SLA

Some service level agreements are enforceable as contracts, but most are agreements or contracts that are more in line with an operating level agreement (OLA) and may not be constrained by law. It is okay to have a lawyer review documents before making any major settlement with a cloud service provider. Service level agreements usually specify certain parameters, which are mentioned below:

- Availability of the Service (uptime)
- Latency or the response time
- Service component's reliability
- Each party accountability
- Warranties

If a cloud service provider fails to meet the specified targets of the minimum, the provider will have to pay apenalty to the cloud service consumer as per the agreement. So, service level agreements are like insurance policies in which the corporation has to pay as per the agreement if an accident occurs.

Big Data

Big Data is a collection of data [both structured and unstructured] that is huge in volume, yet growingexponentially with time.
Examples:-

1. The statistic shows that ***500+terabytes*** of new data get ingested into the databases of social media site **Facebook**, every day. This data is mainly generated in terms of photo and video uploads, message exchanges, putting comments etc.
2. A single **Jet engine** can generate ***10+terabytes*** of data in ***30 minutes*** of flight time. With manythousand flights per day, generation of data reaches up to many ***Petabytes.***

Four Vs of Big Data/ Characteristics Of Big Data

Big data can be described by the following characteristics:

- Volume
- Variety
- Velocity
- Variability

(i) Volume **–** The name Big Data itself is related to a size which is enormous. Size of data plays a verycrucial role in determining value out of data.

(ii) Variety **–** Variety refers to heterogeneous sources and the nature of data, both structured and unstructured.

(iii) Velocity **–** The term **'velocity'** refers to the speed of generation of data. How fast the data is generatedand processed to meet the demands, determines real potential in the data.

(iv) Variability **–** This refers to the inconsistency which can be shown by the data at times, thus hamperingthe process of being able to handle and manage the data effectively.

Sources of Data for Bigdata

A significant part of big data is generated from three primary resources:

- Machine data
- Social data, and
- Transactional data.

[Refer below exam answer]

1. ***Data collected from social Media sites*** like Facebook, WhatsApp, Twitter, YouTube, Instagram, etc
2. ***Sensor placed in various places*** of the city that gathers data on temperature, humidity, etc. A camera placed beside the road gather information about traffic condition and creates data. Security cameras placed in sensitive areas like airports, railway stations, and shopping malls create a lot of data.
3. ***IoT Appliance:*** Electronic devices that are connected to the internet create data for their smart functionality, examples are a smart TV, smart washing machine, smart coffee machine, smart AC, etc. It is machine-generated data that are created by sensors kept in various devices
4. ***Customer feedback*** on the product or service of the various company on their website creates data. For Example, retail commercial sites like Amazon, Walmart, Flipkart, and Myntra gather customer feedback on the quality of their product and delivery time.
5. ***E-commerce:*** In e-commerce transactions, business transactions, banking, and the stock market, lots of records stored are considered one of the sources of big data. Payments through credit cards, debit cards, or other electronic ways, all are kept recorded as data.
6. ***Global Positioning System*** (GPS): GPS in the vehicle helps in monitoring the movement of the vehicle to shorten the path to a destination to cut fuel, and time consumption. This system creates huge data on vehicle position and movement.
7. ***Transactional Data:*** Transactional data, as the name implies, is information obtained through online and offline transactions at various points of sale. The data contains important information about transactions, such as the date and time of the transaction, the location where it took place etc

Role of Big Data in AI & ML

1. Big Data helps machine learning by providing a variety of data so machines can learn more ormultiple samples or training data.
2. In such ways, businesses can accomplish their dreams and get the benefit of big data using MLalgorithms.
3. The larger the amount of data that Artificial Intelligence systems can access, the more machines canlearn and therefore more accurate and efficient their results will be.
4. As AI becomes smarter, less human intervention is required when it comes to process control andmachine monitoring.

Artificial Intelligence applied to Big Data provides the following benefits:

- Deviation detection: AI can analyse the data provided by Big Data to detect unusual occurrencesin it.
- Probability of future outcome: AI can use a known condition with an X probability of influencing the future outcome to determine the probability of that outcome.
- Pattern recognition: Detect patterns from large data structures that humans would be unable to recognize.